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Anomaly Detection: Can you answer these 20 questions?

Machine Learning Interview Questions Part 17

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Crack Data Scientist & Machine Learning Engineer Interview: 400 Questions & Solutions

Anomaly detection is the practice of identifying patterns in data that deviate from expected behavior. In an era where systems generate massive volumes of real-time data, detecting anomalies early is critical for reliability, security, and performance. From spotting fraudulent transactions and network intrusions to detecting equipment failures and data quality issues, anomaly detection plays a central role across industries. Whether you work with time-series data, logs, or complex multivariate signals, understanding anomaly detection is essential for building robust, data-driven systems.



Source: Image is generated by ChatGPT

Let's check your basic knowledge of Anomaly Detection. Here are 20 multiple-choice questions for you.

1. A credit card company uses a model to monitor transactions. A customer who usually spends 50 dollars at local shops suddenly makes a 5,000 dollar purchase in a different country. How would an unsupervised model likely categorize this?

- (A) As a collective anomaly because the single transaction represents a sequence of related events that only become suspicious when viewed as a group over time.
- (B) As a point anomaly because the transaction amount and location significantly deviate from the historical norm for this specific user profile.
- (C) As a contextual anomaly because the transaction would be considered normal if it occurred during a holiday season when high spending is expected.
- (D) As a noise component because machine learning models typically ignore high-value transactions to avoid generating false positives.

2. A factory monitors a cooling fan. The fan vibrates at 10Hz during the day

and 5Hz at night. If the fan vibrates at 10Hz at midnight, the system flags it. What type of anomaly is this?

- (A) Point anomaly
- (B) Contextual anomaly
- (C) Collective anomaly
- (D) Feature scaling anomaly

3. An IT department notices a sequence of failed login attempts followed by a successful login and a massive data download. Individually, these actions happen often, but together they are flagged. Why?

- (A) They represent a point anomaly.
- (B) They represent a seasonal anomaly.
- (C) They represent a collective anomaly.
- (D) They represent a global anomaly.

4. You are training a model to detect rare diseases in medical images. You have 10,000 images of healthy lungs and only 5 images of diseased lungs. Which approach is most appropriate?

- (A) Standard binary classification with Accuracy as the primary metric.
- (B) One-Class SVM or Isolation Forest.
- (C) K-Means clustering with K=2.
- (D) Removing the 5 diseased images to make the dataset cleaner.

5. A manufacturer of aircraft engines uses a One-Class SVM to detect potential failures. The model is trained only on data from engines that passed quality control. Why is the One-Class SVM preferred over a standard Two-Class SVM here?

- (A) Because the manufacturer has an equal amount of fail and pass data.
- (B) Because it is impossible or too expensive to obtain a comprehensive set of failure examples for every possible way an engine can fail.
- (C) Because a One-Class SVM is much faster at processing image data than a

standard SVM.

(D) Because Two-Class SVMs are unable to handle numerical sensor data.

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6. An e-commerce site uses an Autoencoder for fraud detection. During testing, a transaction results in a very high reconstruction error. What does this indicate?

- (A) The transaction is very similar to the training data.
- (B) The model has overfitted and needs more layers.
- (C) The transaction is likely an anomaly.
- (D) The model is successfully compressing the transaction data.

7. In a Local Outlier Factor (LOF) algorithm, a data point is found to have a much lower density than its neighbors. What is the conclusion?

- (A) The point is a global anomaly but not a local one.
- (B) The point is part of a dense cluster.
- (C) The point is a local anomaly.
- (D) The point should be used as a centroid.

8. A data scientist is using DBSCAN for anomaly detection in maritime traffic. After running the algorithm, some data points are not assigned to any cluster. These points are:

- (A) Centroids
- (B) Core points
- (C) Noise or anomalies
- (D) Border points

9. Which of the following is a major challenge when using supervised learning for anomaly detection in cybersecurity?

- (A) The model will be too simple to catch complex attacks.
- (B) Class imbalance and the evolution of new, unseen attack types.
- (C) Supervised models cannot process numerical data.
- (D) There is too much labeled anomaly data available.

10. A smart thermometer flags a temperature of 100 degrees Celsius in a living room. If this is due to the sensor being placed next to a boiling kettle, the anomaly is:

- (A) A malicious anomaly.
- (B) A semantic anomaly.
- (C) A sensing error or noise.
- (D) A structural anomaly.

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11. A logistics company uses a sensor to track the temperature of a refrigerated truck. The temperature stays between 2 and 4 degrees Celsius for three days. Suddenly, the temperature spikes to 20 degrees for 10 minutes before returning to 3. How would an Isolation Forest handle this data point?

- (A) It would treat it as a normal point because the 10-minute duration is too short to be significant.
- (B) It would isolate the point quickly using fewer random splits because it is far from the dense cluster of normal readings.
- (C) It would require a higher depth of trees to find this point because the temperature returned to normal quickly.
- (D) It would ignore the point unless the model was specifically trained on 20-degree examples.

12. A security system monitors network traffic volume. Every Monday at 9:00 AM, there is a massive spike in traffic due to automated backups. If the system flags this spike as an attack, what is the most likely cause of this false positive?

- (A) The model is treating a point anomaly as a collective anomaly.
- (B) The model is not accounting for the temporal context of the data.
- (C) The model is experiencing a sensing error due to hardware failure.
- (D) The model has been overtrained on weekend data only.

13. An industrial water pump is monitored by a One-Class SVM. During a period of heavy rain, the pump's pressure readings shift slightly higher than usual but remain steady. The model flags this as an anomaly. Why does this happen?

- (A) The model has identified a global outlier that is far beyond the maximum possible pressure.
- (B) The new data points fall outside the high-density decision boundary learned during normal operation.
- (C) The One-Class SVM is attempting to cluster the rain data into a second category.
- (D) The model is detecting a collective anomaly where pressure and time are unrelated.

14. A bank uses an Autoencoder to monitor ATM transactions. A fraudster performs 50 small withdrawals of \$20 each within one hour. Each individual transaction is normal, but the reconstruction error for the sequence is very high. What does this indicate?

- (A) The model is failing to compress a collective anomaly.
- (B) The model is detecting a point anomaly based on the \$20 amount.
- (C) The model has encountered a sensing error at the ATM.
- (D) The fraudster has successfully bypassed the reconstruction logic.

15. In a dataset of engine vibrations, most engines vibrate between 50–60 Hz. A few engines show vibrations at 55 Hz but with a very different shape or pattern in the frequency spectrum. Which algorithm is best for finding these outliers?

- (A) Z-Score analysis on the frequency value.
- (B) Simple thresholding at 65 Hz.
- (C) Local Outlier Factor (LOF).
- (D) Linear Regression.

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16. A social media platform wants to detect bot accounts. Bots often follow

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- (B) They are labeled as noise points because their behavior is isolated from the dense clusters of human behavior.
- (C) They are placed in the largest cluster because their activity is very high
- (D) They are used as the epsilon parameter to define the cluster size.

17. A data scientist is using K-Nearest Neighbors (KNN) for anomaly detection. A data point is flagged because its average distance to its 5 nearest neighbors is extremely large. What does this imply?

- (A) The point is in a very dense part of the dataset.
- (B) The point is a global anomaly.
- (C) The point is the centroid of a new cluster.
- (D) The model has too many neighbors (K is too high).

18. An AI system monitors a power grid. It detects a simultaneous drop in voltage across 10 different substations that are not geographically close. No single drop is critical, but the pattern is unique. This is a:

- (A) Point anomaly
- (B) Collective anomaly
- (C) Sensing error
- (D) Noise anomaly

19. You are using an Elliptic Envelope model to detect outliers in a dataset that follows a Gaussian (Normal) distribution. How does this model define what is anomalous?

- (A) It isolates points using a tree structure.
- (B) It fits an ellipse to the central data points and flags anything outside the ellipse.
- (C) It calculates the reconstruction error of a neural network.
- (D) It groups points based on their local density.

20. A cybersecurity firm uses a Gated Recurrent Unit (GRU) network, a type of RNN, to monitor system logs. The model predicts the next log entry based on previous ones. If the actual log entry differs significantly from the predicted one, it is flagged. This is an example of:

- (A) Static point detection
- (B) Time-series forecasting for anomaly detection
- (C) Density-based clustering
- (D) Dimensionality reduction

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Happy learning! Feel free to discuss your thoughts on these questions in the comment section. Don't forget to **clap** and **share** the quiz link with your friends & LinkedIn connections.

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